

Technology Trends & Analysis

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OUTLINE



- Executive Summary
- Introduction
- Methodology
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- Discussion
 - Findings & Implications
- Conclusion
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EXECUTIVE SUMMARY

- This project explores global developer trends using survey data visualized in IBM Cognos Analytics.
- The analysis focuses on key technology and demographic areas, including:
 - Programming languages
 - Databases
 - Development platforms
 - Web frameworks
 - Developer demographics

Key Findings

- **JavaScript & Python:** Most used and high future interest.
- **C#:** Widely used but slower future growth.
- **PostgreSQL & SQL Server:** Leading current and future database choice.
- **Cloud Platforms (AWS & Azure):** Most preferred environments.



INTRODUCTION

Purpose of the Report

To explore and highlight both current and upcoming technology trends in the global developer community.

Target Audience

- Aspiring software developers
- Data analytics professionals
- Technology leaders
- Educational institutions

Values

Provides insights to support **data-driven decisions** in areas such as:

- Career skill development and planning
- Designing relevant educational curricula
- Guiding technology adoption and investment strategies



METHODOLOGY

Data Source

Global developer survey dataset (Stack Overflow–style survey data)

Data Collection

Survey responses were collected from developers worldwide, capturing information about their current technology usage, future interests, and demographic details.

Data Wrangling Steps

- Removed null, duplicate, and inconsistent entries to ensure data quality.
- Standardized technology names to maintain consistency across survey responses.
- Filtered and ranked the top 10 technologies based on response frequency.
- Aggregated counts for:
 - ✓ Current technology usage
 - ✓ Future technology interest

Tools Used

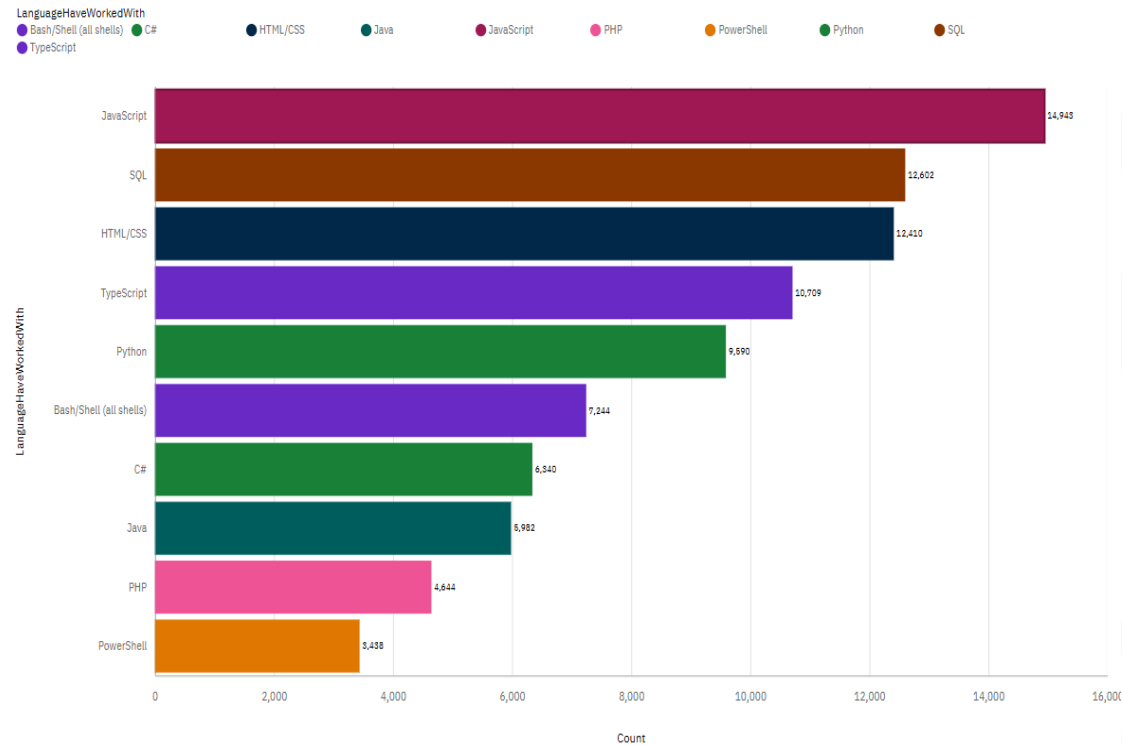
IBM Cognos Analytics for data visualization and dashboard creation



PROGRAMMING LANGUAGE TRENDS

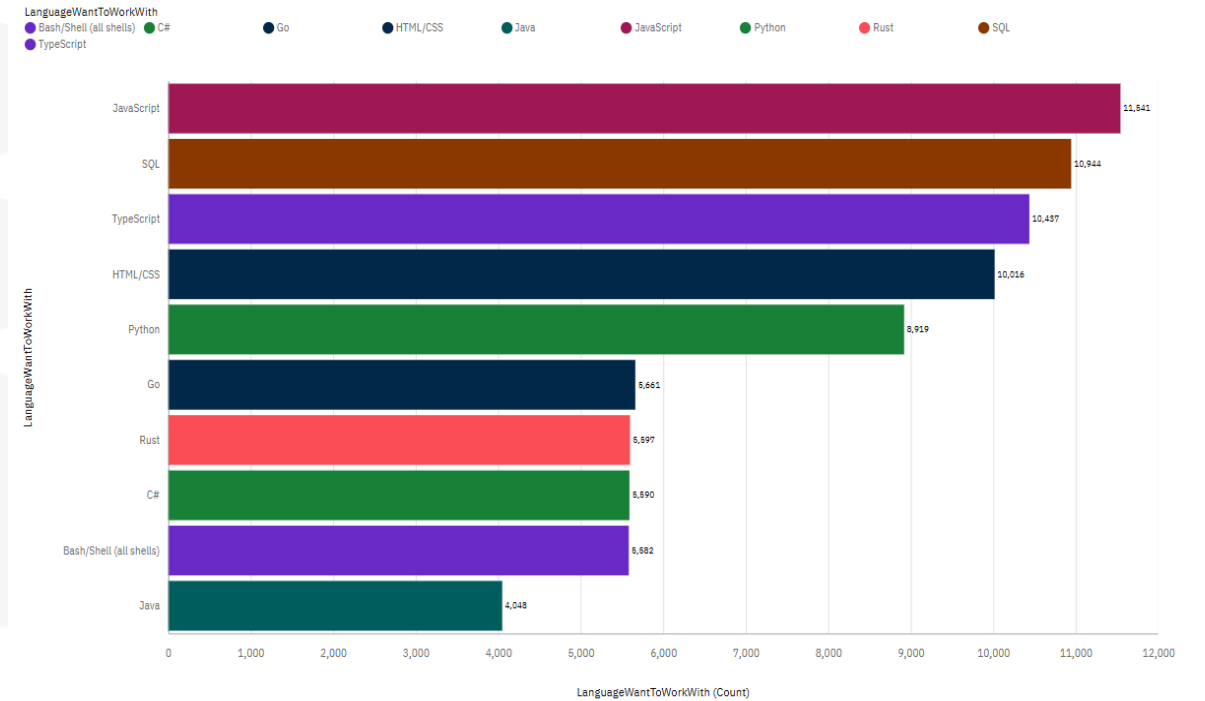
Current Year

Top 10 LanguageHaveWorkedWith



Next Year

Top 10 LanguageWantToWorkWith



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, and HTML/CSS remain foundational for web and data development.
- Python shows strong future interest, driven by AI, data science, and automation.
- TypeScript is gaining popularity as teams adopt scalable and maintainable code.
- Rust and Go appear more frequently in “want to work with,” highlighting emerging demand and skill gaps.
- Traditional languages (Java, C#) are still widely used but exhibit slower projected growth.

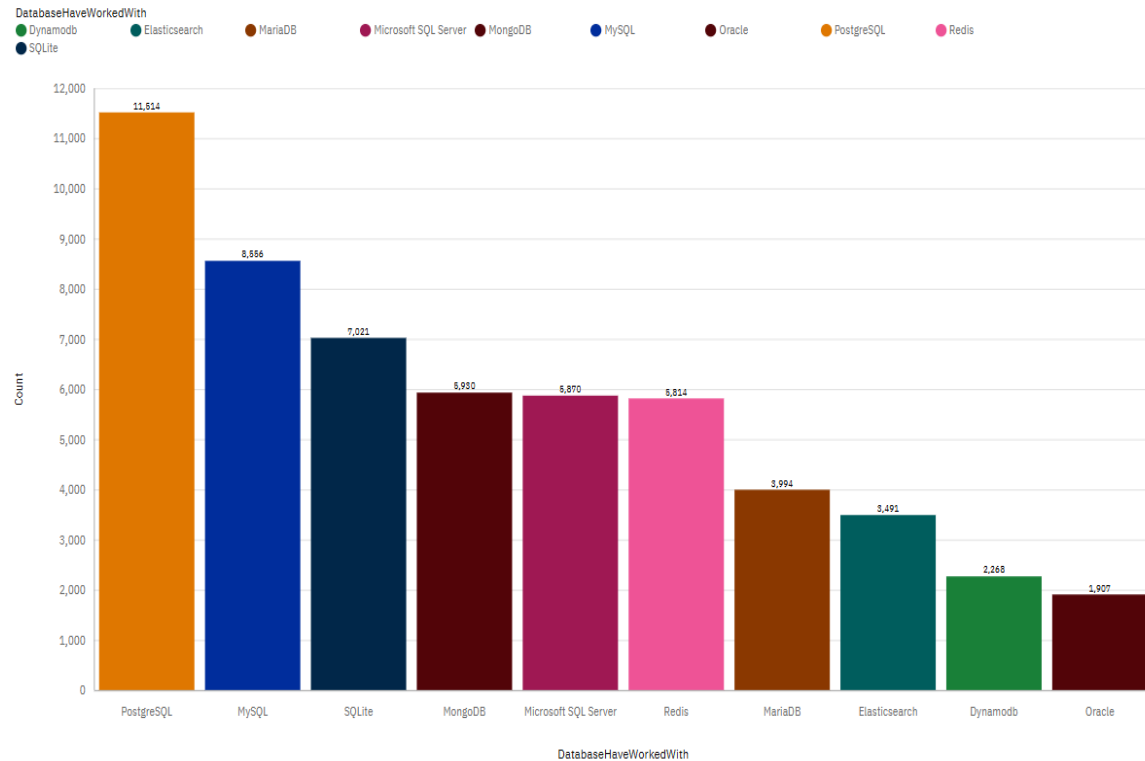
Implications

- Organizations should modernize tech stacks while retaining core languages.
- Upskilling in Python, TypeScript, Go, and Rust is critical for future readiness.
- Hiring and educational strategies should align with cloud-native and data-driven development trends.

DATABASE TRENDS

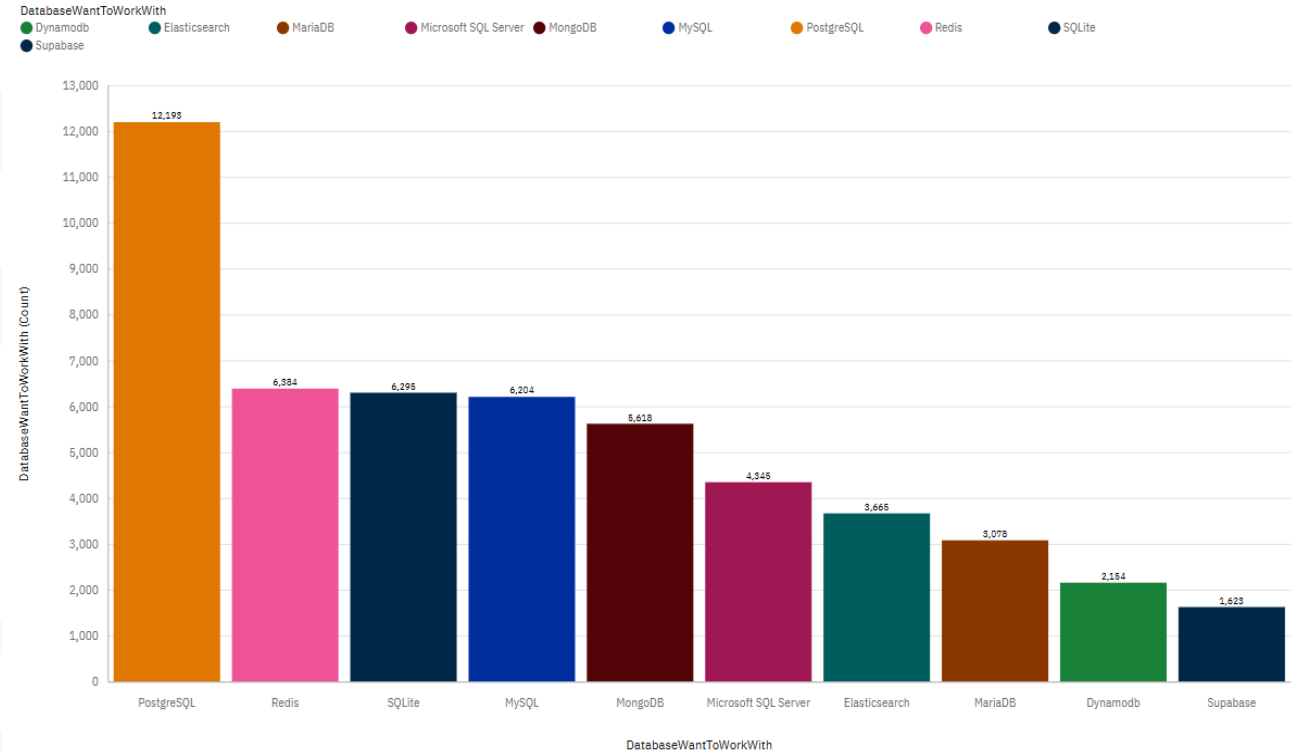
Current Year

Top 10 DatabaseHaveWorkedWith



Next Year

Top 10 DatabaseWantedToWorkWith



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Current Year – Key Findings

- PostgreSQL and Microsoft SQL Server are among the most widely used databases.
- MySQL and MongoDB continue to maintain strong adoption.
- Both relational and NoSQL databases remain relevant in current technology stacks.

Next Year – Key Findings

- PostgreSQL leads in future interest among developers.
- Cloud-compatible and scalable databases are attracting increasing attention.
- Redis and MongoDB reflect growing interest in performance-focused solutions.

Current Year – Implications

- SQL-based databases remain foundational despite the growth of NoSQL.
- Ongoing adoption indicates continued reliance on relational database systems for enterprise applications.

Next Year – Implications

- Organizations should prepare for wider adoption of cloud-native databases.
- Developers should strengthen skills in PostgreSQL and NoSQL databases to stay competitive.

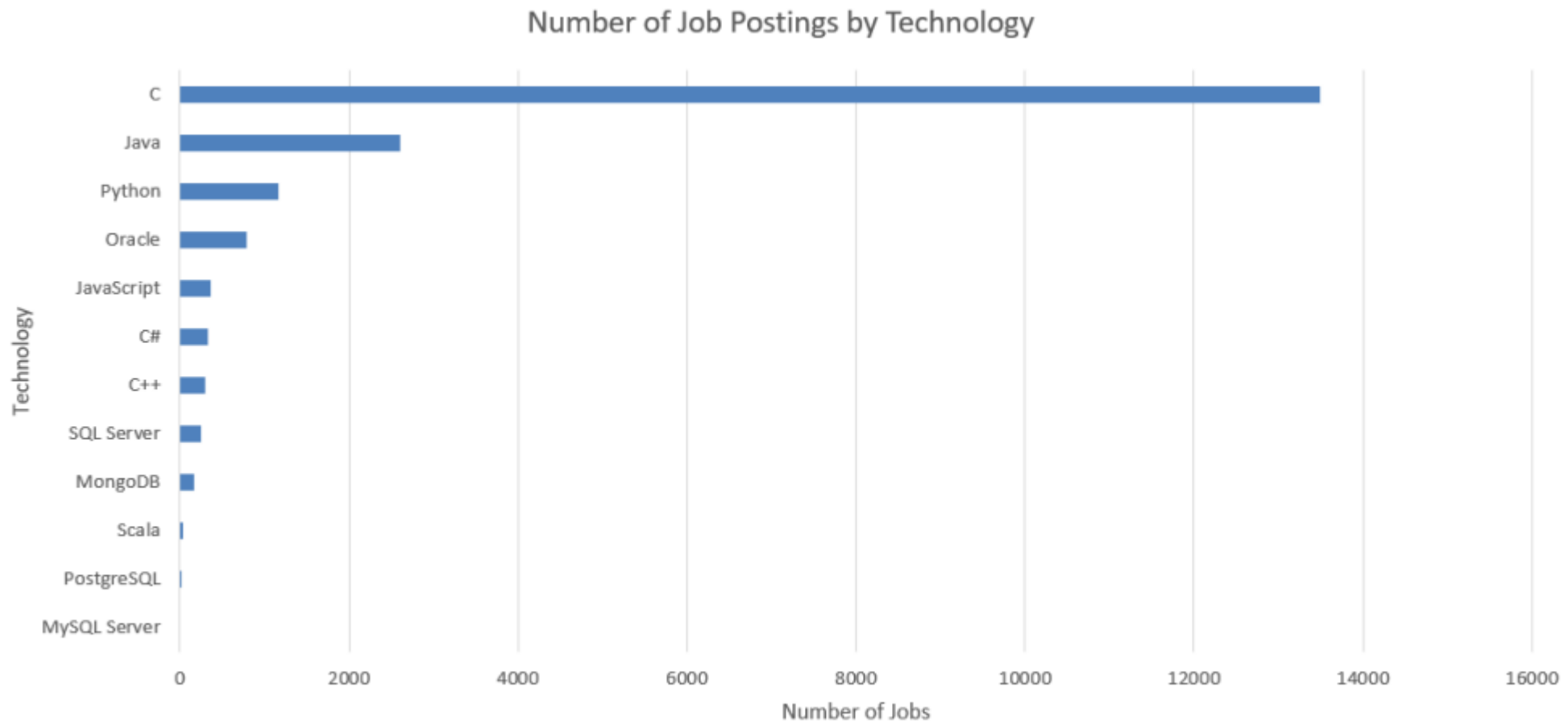


JOB MARKET TRENDS

The following slides present a visual analysis of job postings, highlighting current industry demands and trends



JOB POSTINGS



DASHBOARD



The following slides provide an overview of the dashboard and its insights.



DASHBOARD TAB 1

Current Technology Usage

Future Technology Trend

Demographics

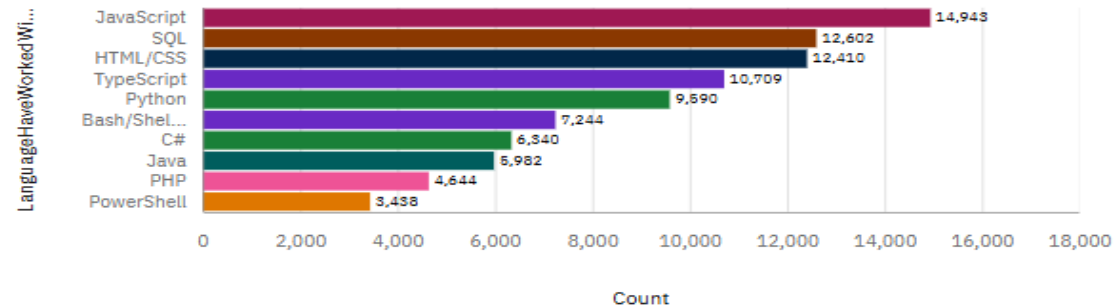
+

Top 10 LanguageHaveWorkedWith



LanguageHaveWorkedWith

Legend: Bash/Shell (all shells), C#, HTML/CSS, Java, JavaScript, PHP, PowerShell, Python, SQL, TypeScript

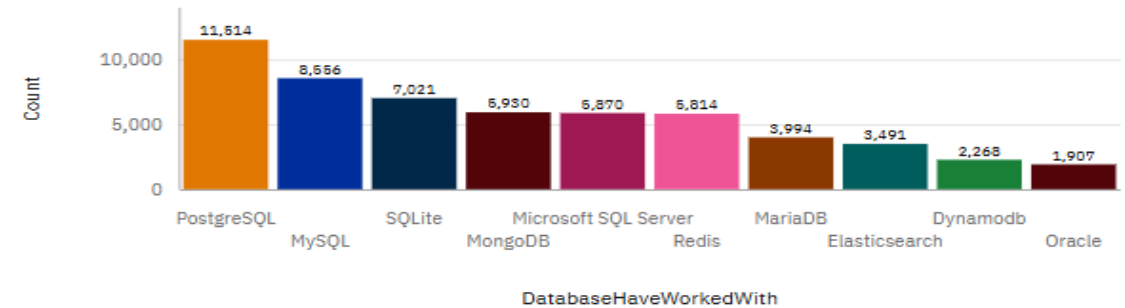


Top 10 DatabaseHaveWorkedWith



DatabaseHaveWorkedWith

Legend: Dynamodb, Elasticsearch, MariaDB, Microsoft SQL Server, MongoDB, MySQL, Oracle, PostgreSQL, Redis, SQLite



Top 10 PlatformHaveWorkedWith



PlatformHaveWor...

Legend: Amazon Web Services (AWS), Cloudflare, Digital Ocean, Google Cloud, Heroku, Hetzner, Microsoft Azure, Firebase

1,392 10,871



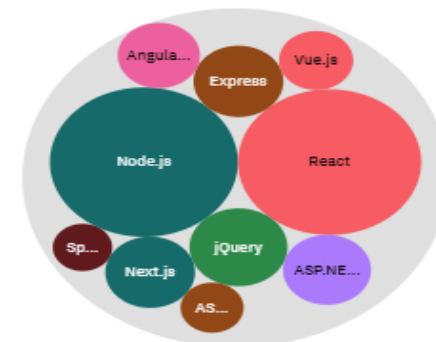
Top 10 WebFrameHaveWorkedWith



WebframeHaveW...

Legend: ASP.NET, ASP.NET CORE, Angular, Express, Next.js, Node.js, React, Spring Boot, Vue.js, jQuery

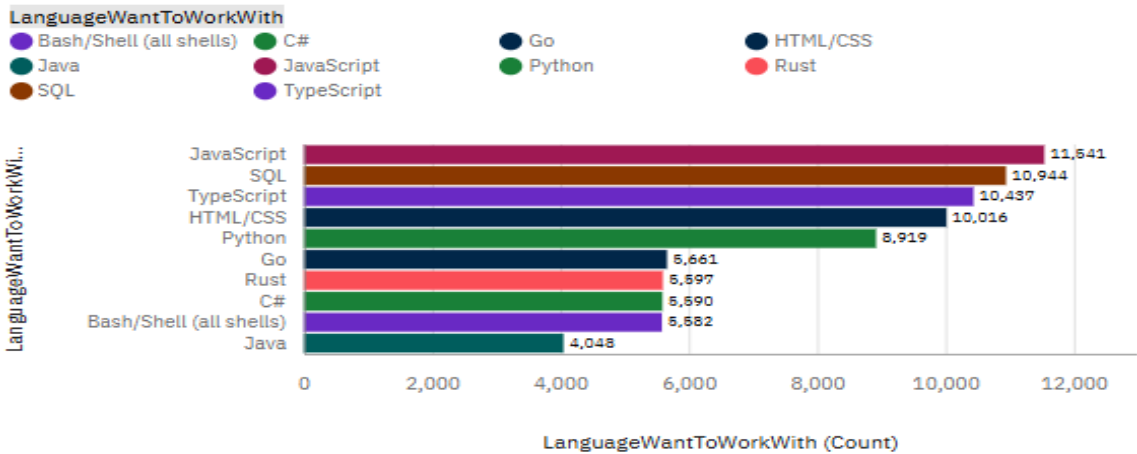
2,916 9,230



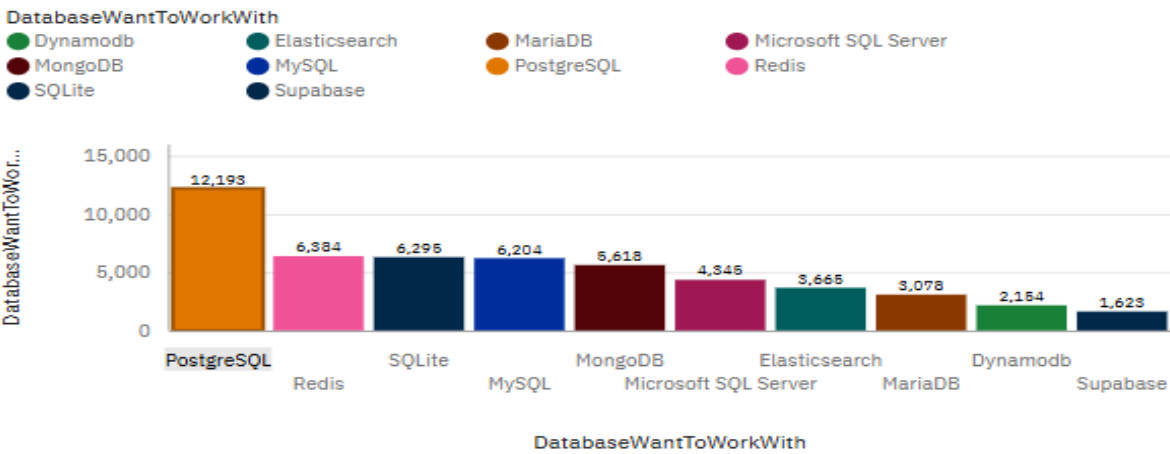
DASHBOARD TAB 2

Current Technology Usage Future Technology Trend Demographics +

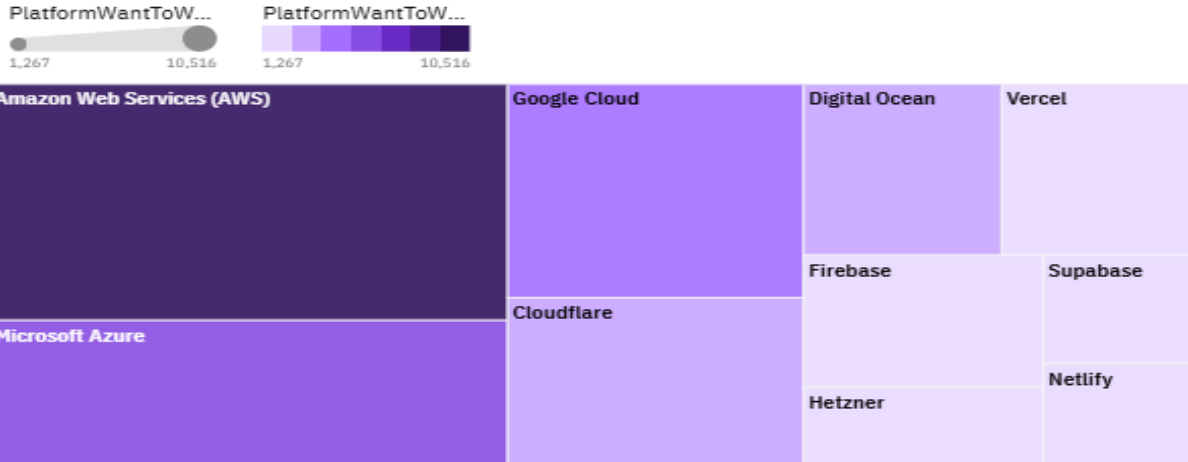
Top 10 LanguageWantToWorkWith



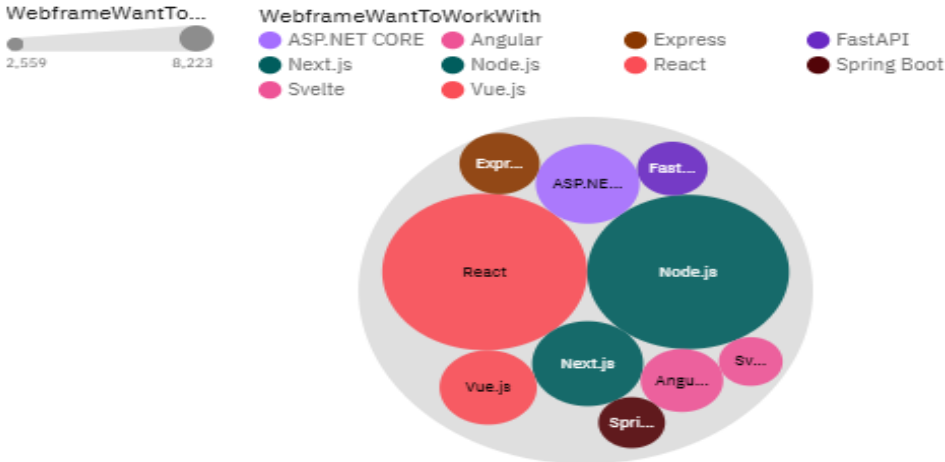
Top 10 DatabaseWantToWorkWith



Top 10 PlatformWantToWorkWith



Top 10 WebframeWantToWorkWith



DASHBOARD TAB 3

Current Technology Usage

Future Technology Trend

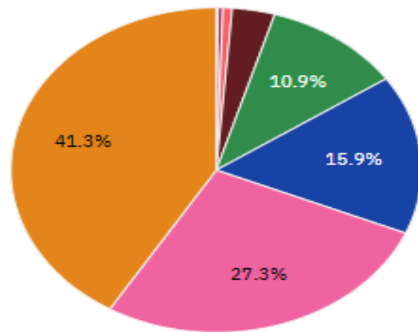
Demographics

+

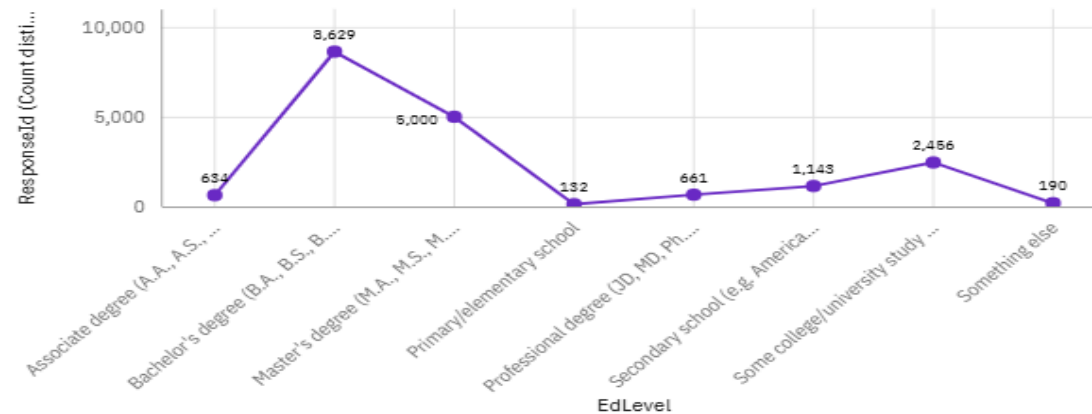
Respondent distribution by Age

Age

- Prefer not to say
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older
- Under 18 years old



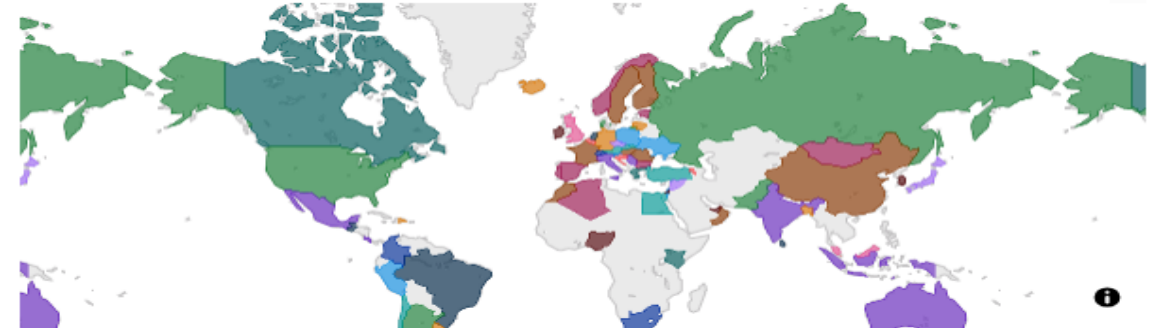
Respondent distribution by Formal Education Level



Respondent Count by Country

ResponseId - Country

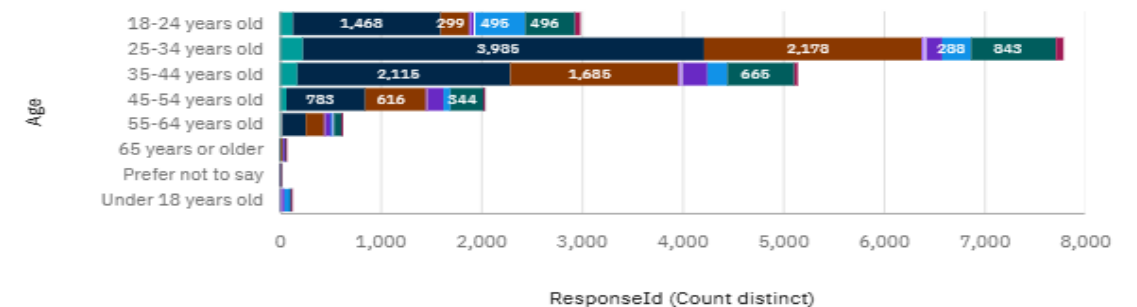
- 3 | United Kingdom of Great Britain...
- 29 | United Kingdom of Great Britain...
- 37 | Austria
- 41 | Brazil
- 45 | Romania
- 122 | Ireland
- 125 | Japan
- 134 | United Kingdom of Great Britain...
- 140 | Pakistan
- 158 | United States of America
- 166 | United States of America
- 182 | Poland



Respondent Count by Age, classified by Education Level

EdLevel

- Associate degree (A.A., A.S., etc.)
- Primary/elementary school
- Some college/university study without degree
- Bachelor's degree (B.A., B.S., B.E., etc.)
- Professional degree (JD, MD, Ph.D., etc.)
- Something else
- Master's degree (M.A., M.S., M.Ed., etc.)
- Secondary school (e.g. American high school)



SK

DISCUSSION

- **JavaScript**-based technologies dominate current development.
- **AWS** and **Azure** are the most commonly used cloud platforms.
- There is a strong shift toward modern, cloud-first technologies.
- Frameworks like **React**, **Fast API**, and **Spring Boot** are gaining increased interest.
- Majority of respondents are aged **25–44**.
- Most respondents hold a **Bachelor's** or **Master's** degree.
- Survey participation is globally distributed.



OVERALL FINDINGS & IMPLICATIONS

Key Findings

- JavaScript, Python, and SQL remain core skills among developers.
- Cloud computing skills are increasingly essential for modern development.
- PostgreSQL is emerging as a preferred and future-proof database technology.

Broader Implications

- Developers should prioritize cloud-native and full-stack skill development to stay competitive.
- Organizations should align hiring strategies with emerging technology trends to attract and retain talent.
- Educational institutions should update curricula to reflect the evolving demands of the developer ecosystem.



CONCLUSION



- Technology is changing gradually, not suddenly.
- Cloud, data, and full-stack development will be important in the future.
- Understanding data can help reduce skill gaps.
- Cognos dashboards effectively support strategic decision-making.

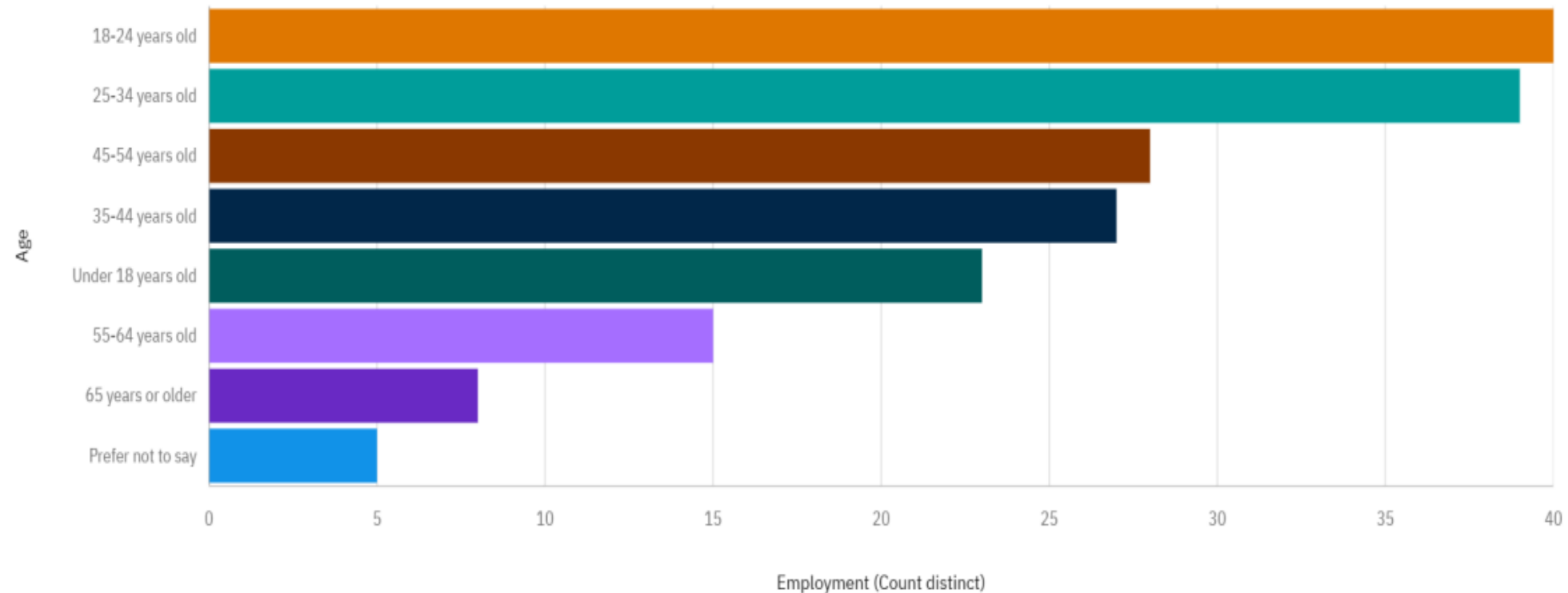


APPENDIX - AGE GROUP V/S EMPLOYMENT STATUS

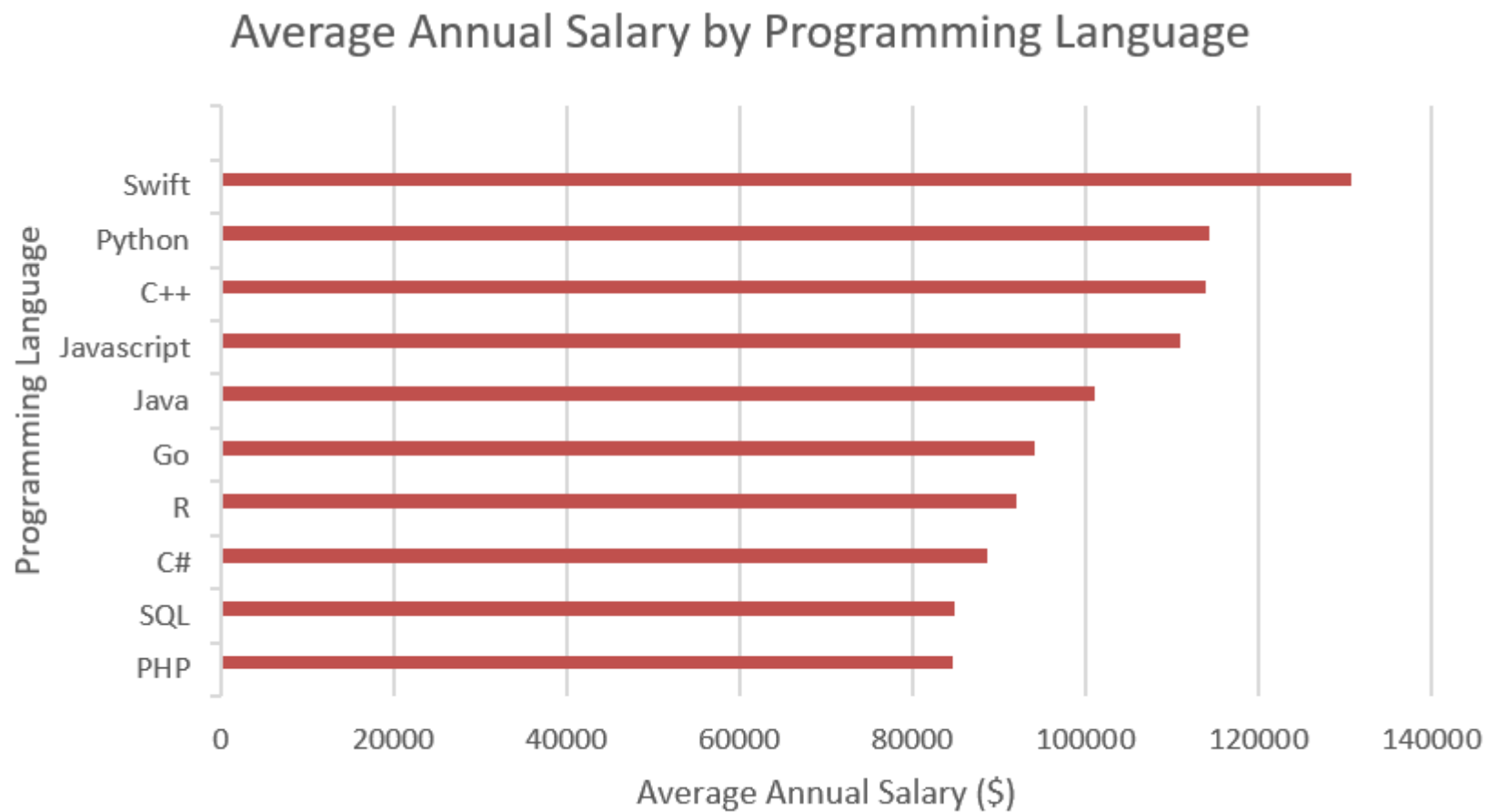
Employment Status Across Age Groups

Age

- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older
- Prefer not to say
- Under 18 years old



POPULAR LANGUAGES





THANK YOU

